



Fig. 2: Accuracy keypad



Fig. 3: Auxiliary keypad



Fig. 4: Base keypad

is very easy to understand and the calculator is accurate up to 16 decimal places.

The program uses some special keys for specific purposes, like 'time' menu to display time, 'MS' for memory storage and 'MR' for memory retrieval. It can also capture the last calculation and use it for consecutive calculations.

The memory storage function uses the hard-disk space of the system. In order to save the data loss, copy the cookies folder in a location other than the main partition that holds the operating system. Cookies are active until one year unless any other data is overwritten.

Note that you have to allow the active contents of the browser before running calculator1.htm program.

Program description

The calculator performs nine types of functions: accuracy, auxiliary, base, Boolean/bitwise, constants, conversions, scientific, standard and statistics.

Accuracy functions. Accuracy functions are used to calculate accurate places of decimal values. Under this function, you have scientific/exponential accuracy, fixed accuracy and precise accuracy modes. Let us consider a number 12.34 up to 3-digit accuracy. For 'scientific' mode, the output will be 1.234×10^1 , 'fixed' mode output will be 12.340 and 'precise' mode output will be 12.3.



Fig. 5: Boolean/bitwise keypad



Fig. 6: Scientific constants keypad

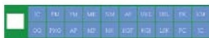


Fig. 7: Scientific conversions keypad

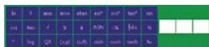


Fig. 8: Scientific functions keypad

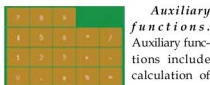


Fig. 9: Standard functions keypad

Auxiliary functions.

Auxiliary functions include calculation of body mass index and simple interest.

Base function. This particular function is mainly used to convert the numbers from one system into another equivalent system. It is divided into two parts:

1. Decimal to any base between 2 and 36
2. Any base between 2 and 36 to decimal

Let us consider decimal number 406501. Converting it into a number of base 4 will give 1203033211. In the same manner, EFY of base 36 when converted into decimal gives 18718. You can visit www.unitconversion.org website for details of conversion utility.

Boolean/bitwise functions. This particular function performs simple bitwise operations like AND, OR, NOT, XOR, binary left-shift and binary right-shift. The binary left-shift operation converts the input into its equivalent binary digits and then shifts the answer with the provided number of left shifts.

For example, $406501 \text{ AND } 2 = 0$,



Fig. 10: Statistical functions keypad

$406501 \text{ OR } 2 = 406503$, $\text{NOT } 406503 = -406502$, $406501 \text{ XOR } 2 = 406503$, 406501 left-shifted by 2 digits = 1626004, 406501 right-shifted by 2 digits = 101625.

Constants. This part of the calculator consists of basic scientific constants (courtesy: CASIO Natural Text Display fx-991 ES) that can be used in various mathematical calculations. A total of 40 keys are provided for the constants (these are mathematical constants from the JavaScript Math library).

Conversions. This part consists of scientific conversions (Courtesy: CASIO Calculator Model: CASIO Natural Text Display fx-991 ES) that can be used in various mathematical calculations. Twenty keys are provided here for various conversion parameters.

Scientific calculator. This part of the calculator performs basic scientific functions like natural logarithm, Bragg's logarithm, factorial, inverse of sine, cosine and tangent (both in degrees as well as in radians). The functions are percentage, Euler's constant, derivative of first kind, discount, integration of first kind, inverse, power, Bragg's logarithm, quadratic roots, polar to rectangular coordinates, rectangular to polar coordinates, hyperbolic sine, hyperbolic cosine and hyperbolic tangent.

Standard. This part has an inbuilt keypad to input the data directly and display the answer. The operators included are add, subtract, multiply, divide and remainder.

Statistics. This part of the calculator can perform basic one-variable statistical data analysis including sum, sum of the squares, mean, variance and standard deviation. It has ten dedicated input cells.

EFY note. The source code of this article has been included in the month's EFY-CD and is also available on www.efymag.com. ●